Cross-Bank Fraud Detection

Group 17.2
supervised by Prof Steve Ling

Shreeshailya Patil—26226492
Ordonez Chala, Nicolas—26041259
Yike Li - 25591266
Tasnuba Sraboni - 25955422

49275 Neural Networks and Fuzzy Logic
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Agenda

1. Introduction
2. Literature Review
3. Aims & Objectives
4. Methods: Approach, Data & Evaluation
5. Preliminary Results
6. Timeline
7. Team Contributions
8. References
9. Q&A

1. Introduction

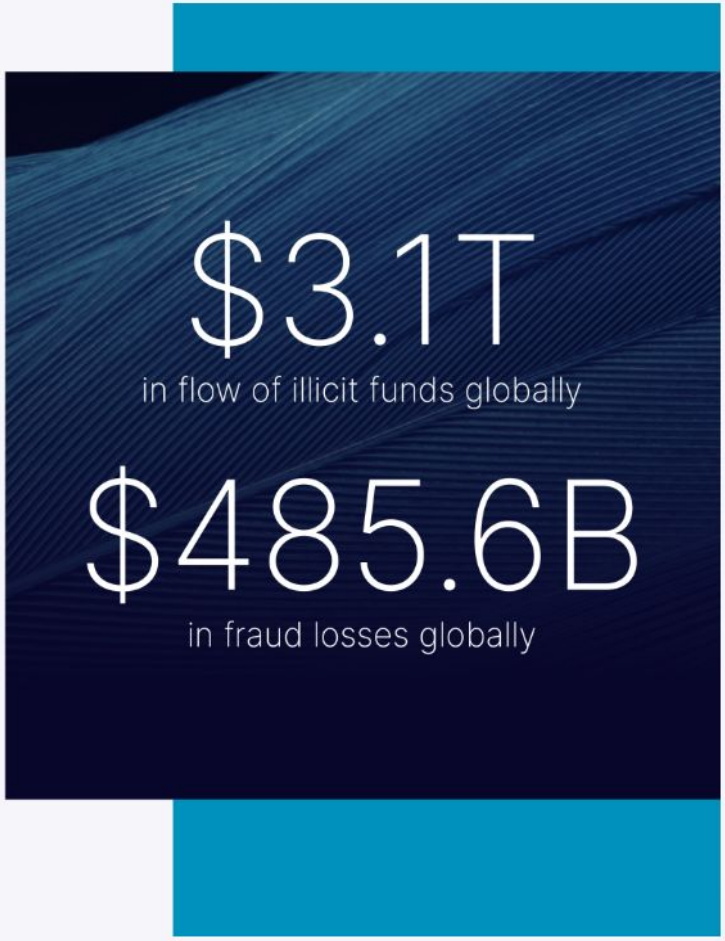
- Overview
- Literature Review
- Preliminary ideas
- Motivation, aims and objectives
- Datasets

Overview

The Global Scale of Financial Crime

In 2023, an estimated $3.1T in illicit funds flowed through the global financial system. Money laundering accounted for trillions of dollars funding a range of destructive crimes, including an estimated $346.7B in human trafficking and $782.9B in drug trafficking activity, as well as $11.5B in terrorist financing.

In 2023, fraud scams and bank fraud schemes totaled $485.6B in projected losses globally.



Literature Review

Following PRISMA Framework from 2020

IDENTIFICATION

Databases: IEEE, ACM, Springer, Elsevier, MDPI
+ Citation screening + Google Scholar alerts

510 records

SCREENING

Title/Abstract review
Excluded: domain mismatch (E4), date range (E2),
no DOI (E3), opinion papers (E5), non-English (E6)

118 records

ELIGIBILITY

Full-text assessment
Excluded: IID-only (E7), non-Q1/Q2 (E8),
no quantitative results (E5), preprint-only (E1)

42 records

INCLUDED

Qualitative synthesis
28 papers from 2025 | 11 papers from 2024
3 foundational baseline references

42 INCLUDED

Research gaps

Multiple Federated Learning (FL) approaches

- FedAvg (2017)
- Personalized Fed 2020
- FedProx (2020)
- SCAFFOLD (2020)
- FedRLCS (2023)
- FedNova (2025)

!

BUT...

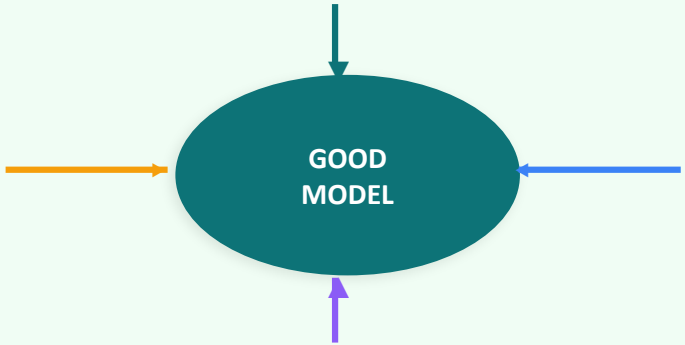
- Unmeasurable metrics across studies
- Different Datasets
- Different goals (credit score, money laundering)

The Non-IID Problem

IID = Identically & Independently Distributed: all banks would have similar fraud patterns.
Reality: every bank is different.

IDEAL

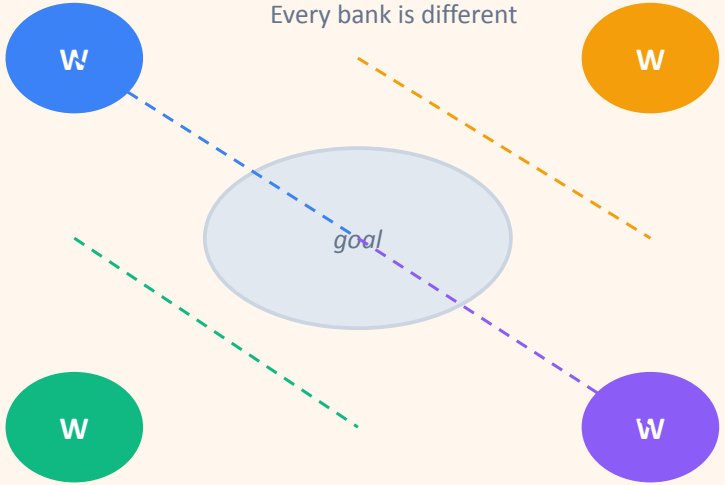
Same data everywhere



All banks agree. ✓

REALITY

Every bank is different



Banks pull in different directions. ✗

Motivation

01

Different algorithms behave differently

02

No standardised benchmark to evaluate their effectiveness for fraud detection

03

Under extreme non-identical (non-IID) data distributions, existing FL methods behave inconsistently

Why Federated Learning?

🔒

Legal Barriers

GDPR & Australian Privacy Principles ban sharing raw transaction data between banks

🏦

Competitive Sensitivity

Banks will not share customer behaviour data with rival institutions

📉

Limited View

A single bank only sees its own fraud patterns—rare events, skewed distributions

The FL Solution

🔗 Each bank trains locally—zero raw data ever leaves
🤝 Only model updates (weights) are shared with a central coordinator
📈 Collaborative learning improves fraud detection for everyone
✅ Scales to real-world multi-bank deployment without regulatory risk

Why Federated Learning?

The core problem: banks cannot share raw transaction data

- Legal barriers—GDPR (Europe) and Australian Privacy Principles prevent centralised data pooling
- Competitive sensitivity—banks will not share customer behavioural data with rivals

Why not a centralized model?

- A single bank sees only its own fraud patterns—rare events, skewed distributions
- Centralised training requires raw data transfer—legally and practically infeasible

Why FL specifically?

- FL trains locally at each bank and shares only updated parameters —zero raw data exposure
- Collaborative learning improves minority-class detection across all participants
- Scales to real-world multi-bank deployment without regulatory risk

Motivations: Aims & Objectives

Main Aim: Benchmark FL algorithm under extreme non-IID conditions for cross-bank fraud detection

01

Standardised Baseline

Establish a common FL algorithm and evaluation protocol that everyone can reproduce

02

Algorithm Comparison

Compare FedAvg, FedProx, SCAFFOLD, and FedNova on the same fraud dataset

03

Performance & Privacy

Evaluate performance adding a privacy layer into the architecture

Expected Outcome

A standardized, reproducible benchmark comparing FL algorithms under extreme non-IID conditions for cross-bank fraud detection.

Ranked Comparison

FedAvg, FedProx, SCAFFOLD & FedNova ranked on unified fraud datasets

Best Algorithm Identified

Winner across three dimensions: detection accuracy, fairness, and privacy

Reusable Protocol

A documented evaluation protocol (metrics + dataset split) for future FL fraud research

Privacy-Utility Guide

Practical guidance on privacy vs. accuracy trade-offs using DP-SGD in federated settings

Method: FL Approach

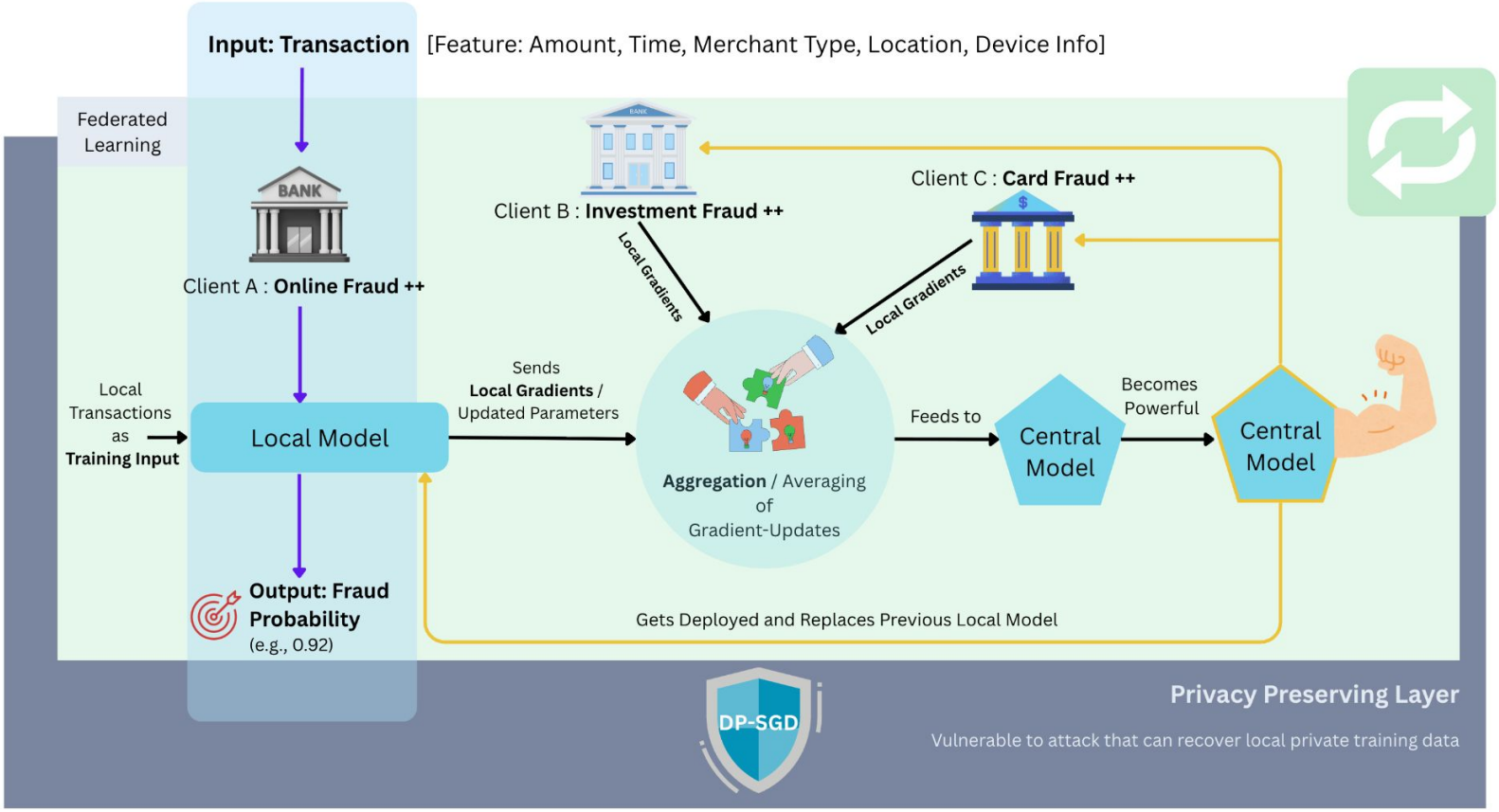


Diagram: Banks collaboratively train a fraud detection model on not independent-and-identically-distributed (non-IID) data without sharing sensitive information

Method: FL Approach

Fraud Training Models

Restriction

Data privacy regulations limit data sharing for training models

- General Data Protection Regulation in Europe
- Australian Privacy Principal acts in Australia

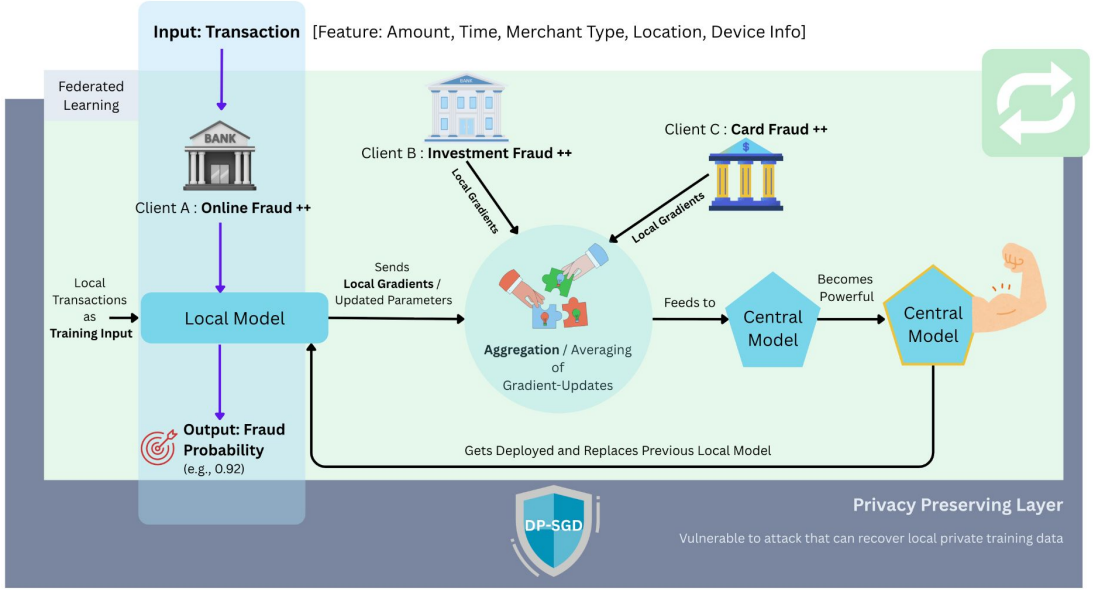
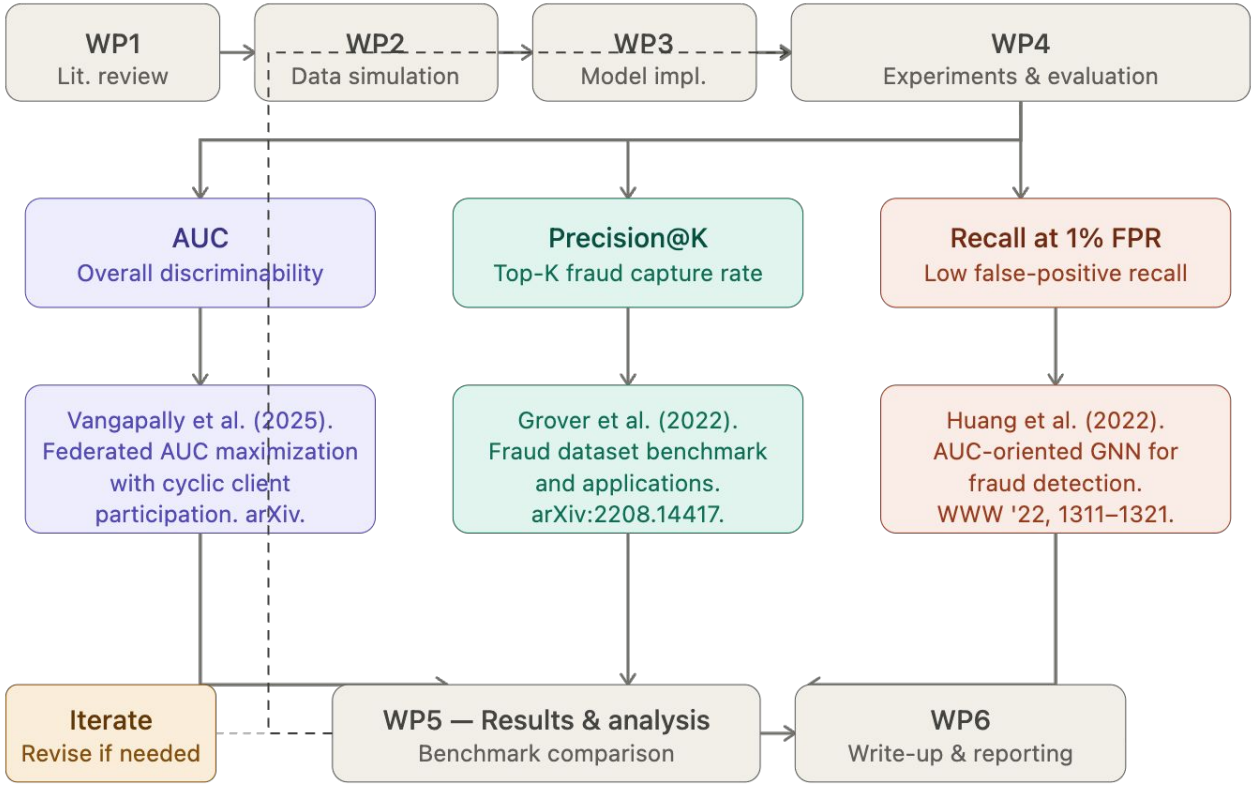


Diagram: Banks collaboratively train a fraud detection model on not independent-and-identically-distributed (non-IID) data without sharing sensitive information

Preliminary Ideas

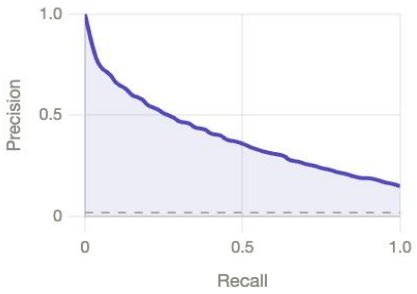


Preliminary Ideas—What might be a better plan?

METRIC 1 — UPGRADED

AUPRC

How well does the model find fraud across all precision–recall tradeoffs? Especially useful when fraud is rare.



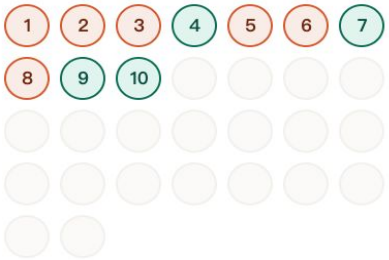
Replaces AUC — why?
ROC–AUC inflates when negatives dominate. AUPRC focuses on the minority class — critical at 0.13%–0.17% fraud rates.

METRIC 2 — UPGRADED

Card Precision@K

Of the top–K flagged cardholders, how many are truly fraudulent? One case per cardholder, not per transaction.

Top K=10 reviewed | **Precision = 70%**



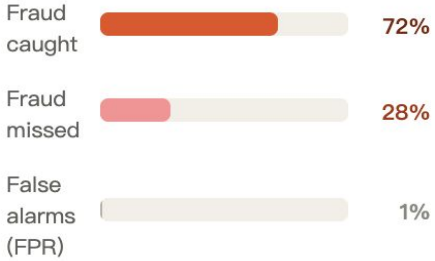
Each circle = one cardholder (not transaction)

Replaces Precision@K — why?
Standard P@K counts transactions — but investigators work by cardholder. Card P@K matches real operational workflow (Lebichot et al., 2022).

METRIC 3 — RETAINED

Recall at 1% FPR

How much real fraud do we catch when we can only afford to falsely flag 1% of legitimate transactions?



FPR is fixed — only recall changes with model quality

Kept as operational baseline
FPR constraint mirrors real bank ops. Pairs with AUPRC to give both global and threshold–specific performance.

Preliminary Ideas
—What might be a better plan?

ORIGINAL PLAN

AUC
Can be misleadingly high under extreme class imbalance (0.17% fraud rate)

Precision@K
K is subjective; treats each transaction equally regardless of cardholder

Recall at 1% FPR
1% FPR threshold is assumed — real banks have different operational constraints

Iterate →

IMPROVED PLAN

AUPRC replaces AUC
Area under Precision–Recall curve — far more sensitive to imbalanced data

Card Precision@K replaces Precision@K
Groups by cardholder — matches real investigator workflow (Lebichot et al., 2022)

Recall at 1% FPR retained
Kept as operational baseline — pairs well with AUPRC for full picture

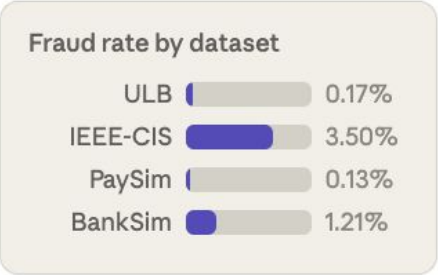
Dataset and Materials

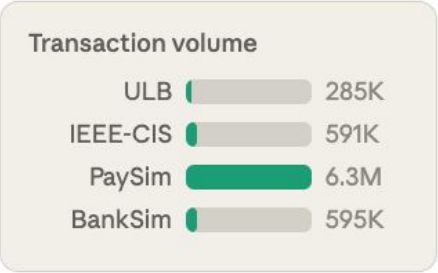
Datasets from: [Workshop on AI for Financial Crime Fight AI4FCF @ ICDM 2025 (IEEE International Conference on Data Mining 2025—Washington DC, USA)]

	DATASET 1 Credit Card (ULB)	DATASET 2 IEEE–CIS	DATASET 3 PaySim	DATASET 4 BankSim
Type	Real	Real	Synthetic	Synthetic
Transactions	284,807	590,540	~6.3M	594,643
Features	30	431	11	10
Fraud rate	0.17%	3.5%	~0.13%	~1.21%
Anonymised	Yes (PCA)	Partial	N/A	N/A
FL role	Fraud detection benchmark	Rich feature heterogeneity	Non–IID simulation base	Agent–based FL simulation

Dataset and Materials

1 Data distribution





2 Preprocessing pipeline

1. Imbalance handling — SMOTE / undersample
2. Feature standardisation — Min-max / z-score
3. Missing value imputation — Median / mode
4. FL partitioning — Geo / income / channel

3 Dataset samples

Credit Card (ULB)

Time	Amount	V1	V2	Label
0	149.62	-1.36	-0.07	Fraud
1	2.69	1.19	0.27	Legit

PaySim

Type	Amount	OldBal	NewBal	Label
TRANSFER	181.00	181.00	0.00	Fraud
PAYMENT	9839.64	170136	160297	Legit

BankSim

Step	Category	Amount	Age	Label
1	leisure	4.55	2	Legit
12	travel	980.00	4	Fraud

Dataset and Materials

LIMITATIONS

Dimension	DATASET 1 Credit Card (ULB)	DATASET 2 IEEE-CIS	DATASET 3 PaySim	DATASET 4 BankSim
BASIC STATISTICS				
Type	Real	Real	Synthetic	Synthetic
Transactions	284,807	590,540	~6.3M	594,643
Features	30	431	11	10
Fraud rate	0.17%	3.50%	0.13%	1.21%
Anonymised	Yes (PCA)	Partial	N/A	N/A
COMPLEMENTARITY				
Controllability	Fixed	Fixed	Tunable	Tunable
FL role	Fraud detection benchmark	Rich feature heterogeneity	Non-IID simulation base	Agent-based FL simulation

strength weakness moderate

WHY FOUR DATASETS?

REAL-WORLD VALIDITY
ULB + IEEE-CIS
Live transaction patterns — ecological validity

CONTROLLED SIMULATION
PaySim + BankSim
Tunable non-IID ground truth for FL benchmarking

SCALE DIVERSITY
285K → 6.3M rows
Tests FL scalability across data volume regimes

FEATURE COMPLEXITY
10 → 431 features
Stress-tests FL across dimensionality extremes

Each weakness on the left is offset by another dataset's strength — **no single dataset covers all dimensions.**

Datasets from: [Workshop on AI for Financial Crime Fight AI4FCF @ ICDM 2025 (IEEE International Conference on Data Mining 2025—Washington DC, USA)]

Evaluation Metrics

Detection Performance

- AUPRC—primary metric; addresses class imbalance in fraud detection
- F1-Score, Precision, Recall—per-client and global

Convergence & Efficiency

- Rounds to convergence, communication cost per round

Fairness & Privacy

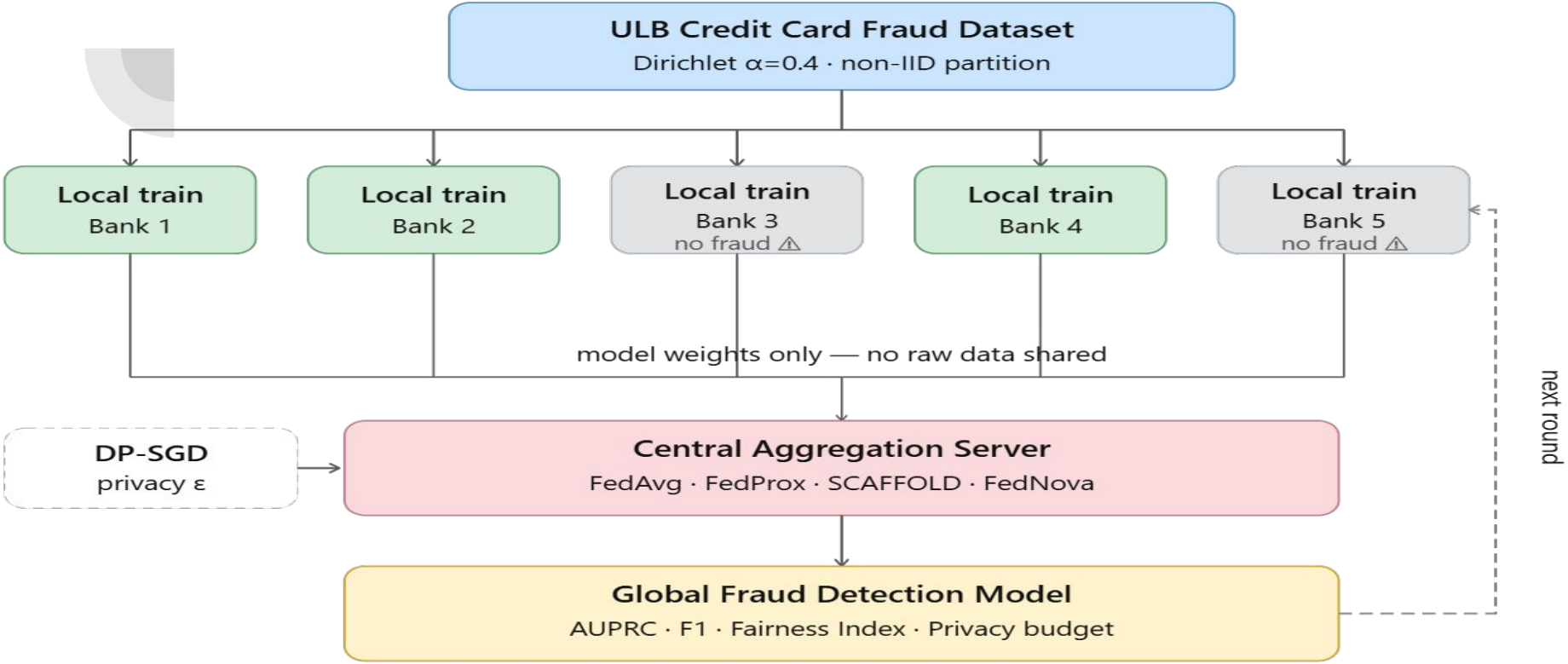
- Jain Fairness Index across banks; DP-SGD privacy budget (ε)

2. Workflow

Overview

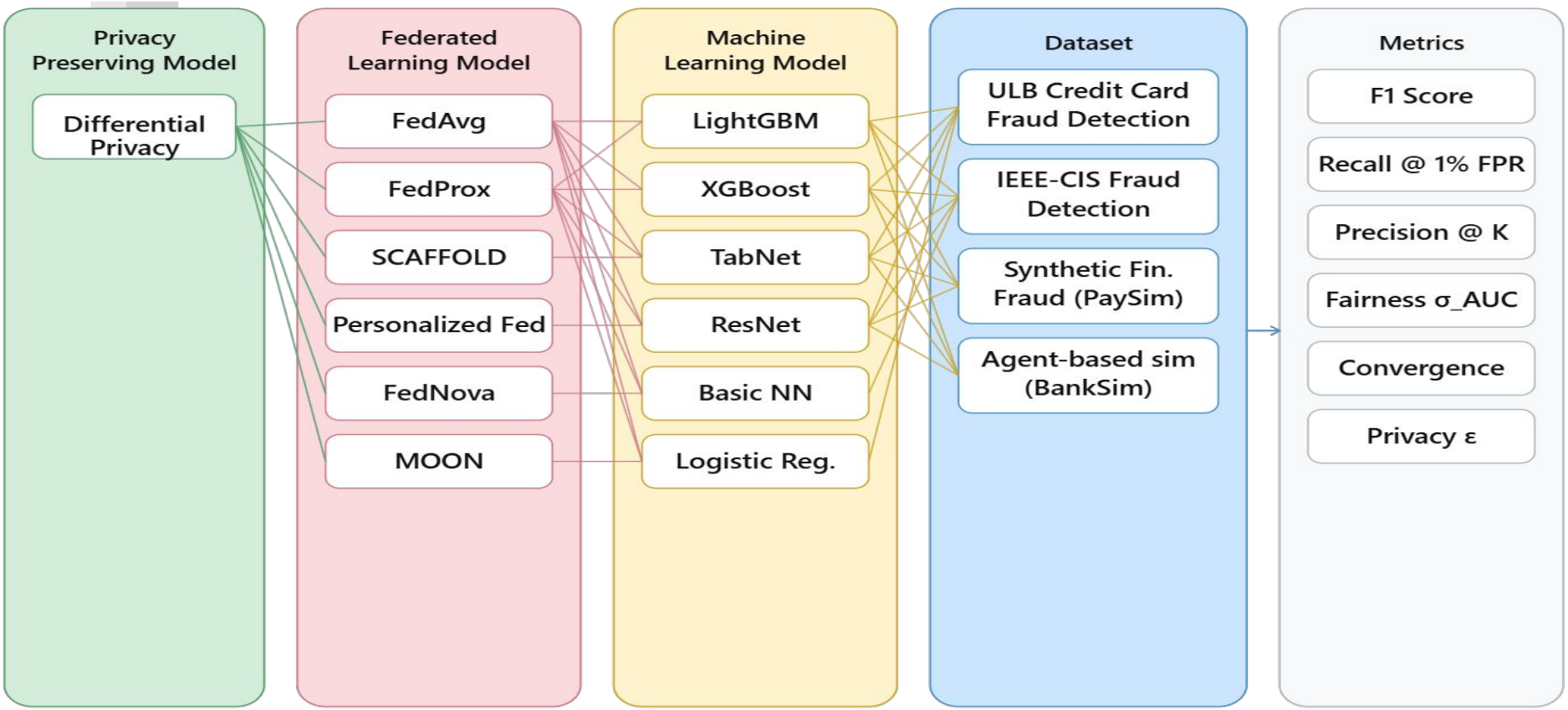
- Methodology
- Current State of Progress
- Team Contributions

Workflow Overview



60 communication rounds · 5 banks · non-IID (Dirichlet α=0.4)

Methodology



Current State of the Project

Completed

- 60+ paper literature survey (PRISMA), dataset acquisition & preprocessing
- FedAvg baseline + FedProx, SCAFFOLD, FedNova implemented and benchmarked
- PersFL+DP, FedAvg+XGBoost, SCAFFOLD+TabNet, FedNova+ResNet variants run

In Progress

- Full fairness analysis across all algorithm variants
- Ablation study on Dirichlet α values (0.1, 0.4, 1.0)
- Paper write-up and Seminar 2 preparation

Current State of the Project

Completed

Literature survey & data preparation
- 60+ papers (PRISMA)
- 4 datasets acquired
- Preprocessing done

Baseline FL algorithms implemented & benchmarked
- FedAvg
- FedProx
- SCAFFOLD
- FedNova

Algorithm variants run
- PersFL+DP
- FedAvg+XGBoost
- SCAFFOLD+TabNet
- FedNova+ResNet

In progress

Full fairness analysis
- All algorithm variants
- Per-client AUC variance

Ablation study on Dirichlet α values
- α = 0.1
- α = 0.4
- α = 1.0

Paper write-up & Seminar 2 preparation
- Results synthesis
- Final report

Overall progress ~65%

Completed In progress

Preliminary Result

Current Progress

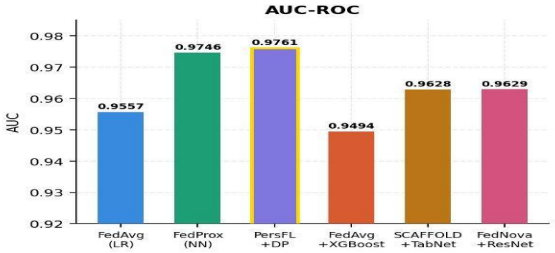
- FedAvg baseline implemented on AI4FCF dataset (GitHub)
- Differential privacy layer (DP-SGD) and Byzantine fault tolerance integrated

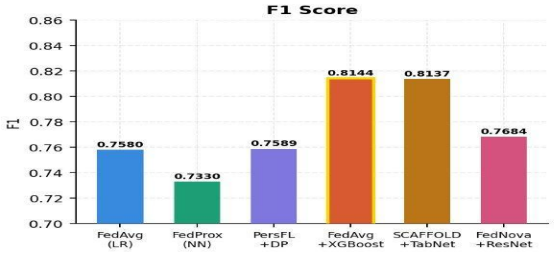
Remaining Work

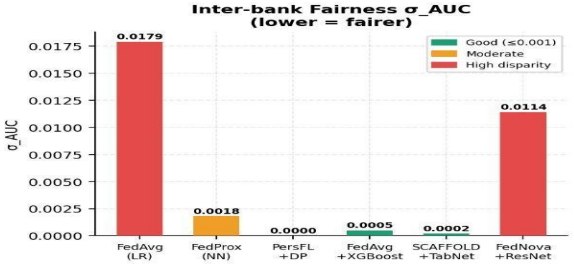
- Run FedProx, SCAFFOLD, FedNova under identical non-IID splits
- Collect AUPRC, F1, fairness index, and privacy budget results for comparison

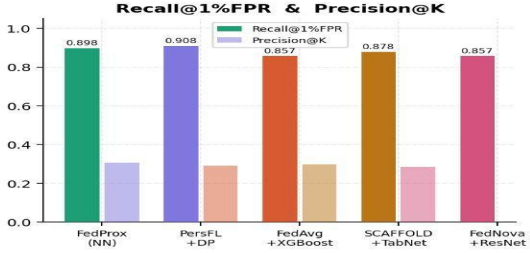
Preliminary Results Benchmark

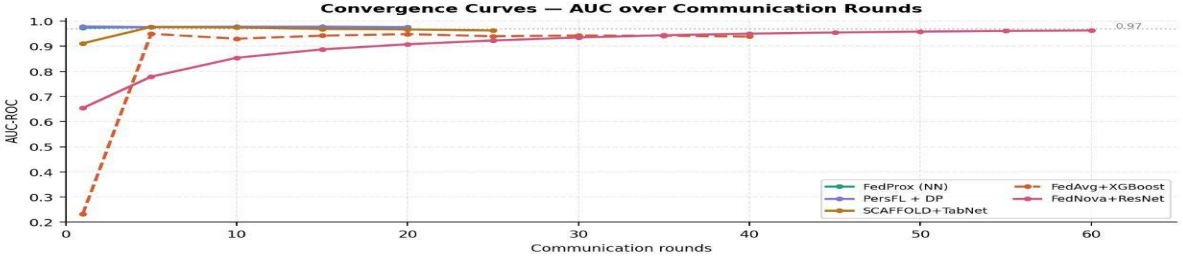
Federated Learning Benchmark — Cross-Bank Fraud Detection
ULB Credit Card Fraud Dataset · 5 Banks · Non-IID (Dirichlet α=0.4)

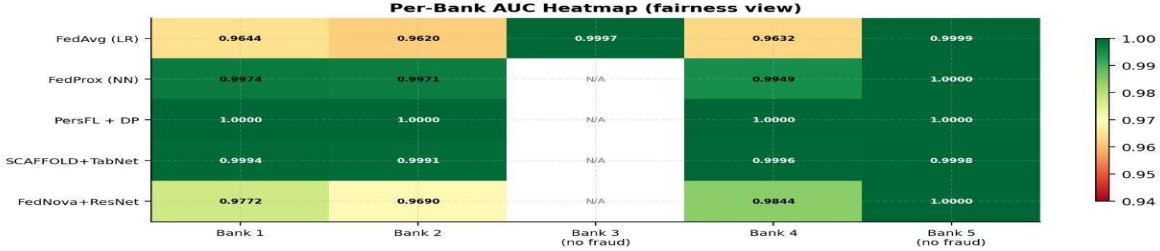


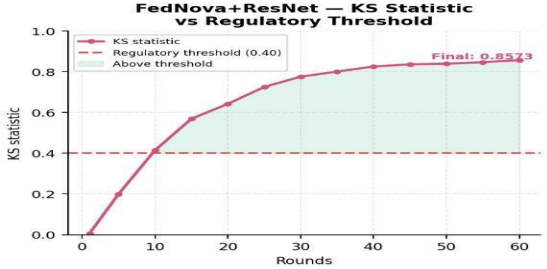












Current State of the Project

- Are there any issues?
- Have you encountered any difficulties?
- Do you have any preliminary results?

Project Timeline

Phase 1 — Weeks 1–4 (Complete)

- Literature review (60+ papers, PRISMA), dataset acquisition, FedAvg baseline

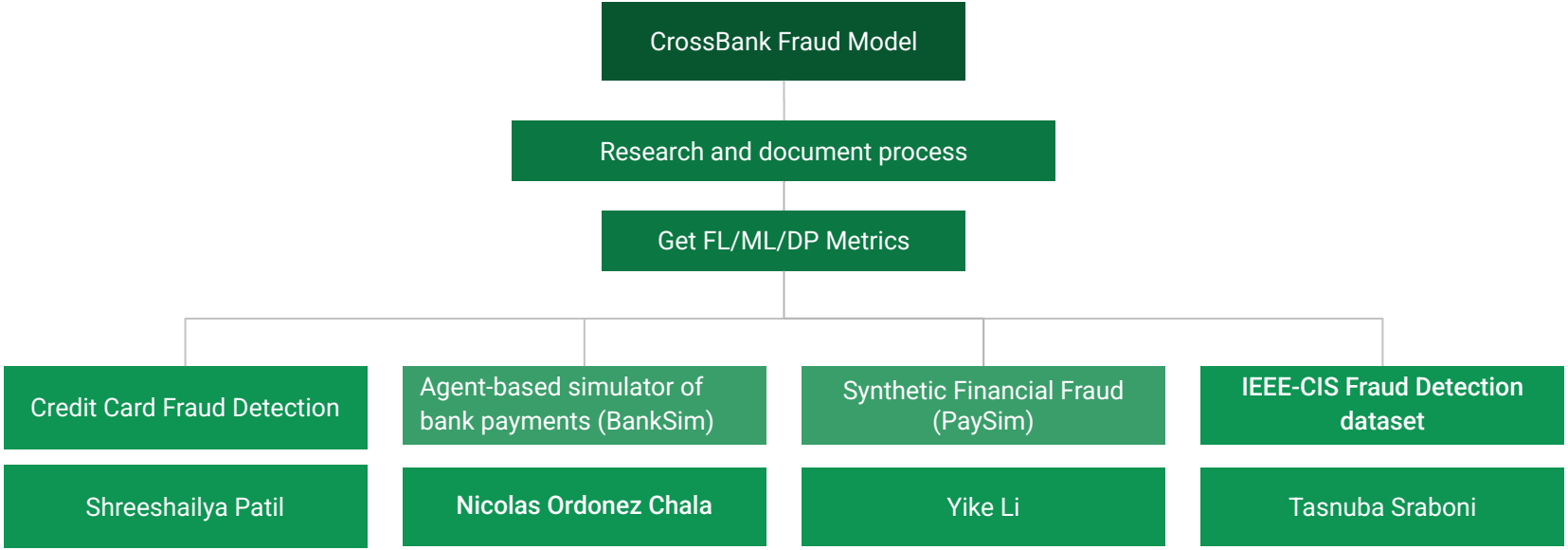
Phase 2 — Weeks 5–9 (In Progress)

- Implement FedProx, SCAFFOLD, FedNova; run controlled non-IID experiments

Phase 3 — Weeks 10–13 (Planned)

- Analysis, fairness/privacy evaluation, paper writing, Seminar 2 presentation

Contribution Statement



Additional notes

- 1. Introduction—Statistical info
- 2. problem statement—existing issue (research gap) , why it’s important
- 3. How AI can benefit
- 4. Aim and objective
- 5. motivation—is to solve "this" problem (connected to research gap)
- 6. Expected outcome—Identify major aim and how to achieve it. (connect to objectives)
- 7. List 2-3 objectives to achieve major aim
- 8. ** Why did you choose this model to solve the problem? Give a reason why “this” idea
- 9. ** What is input (e.g, image) and what is expected output (e.g., image/yes or no)
- 10. Include a block diagram (all the process of 9 (input to output)).
- 11. Dataset overview: Show how many data samples there are, where you get them, and a sample of the data.
- 12. What's the progress? What has is achieved so far
- 13. Contribution: Create a table
- 14. Highlight Key features of the approach (for us it’ll be features of FL)
- 15. Put more images and diagrams and less words
- 16. Reference the images (if taken from any online source)

References

Federated Learning — Core Algorithms

- McMahan, H. B., Moore, E., Ramage, D., Hampson, S., & Arcas, B. A. (2017). Communication-efficient learning of deep networks from decentralized data. *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 54, 1273–1282.
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Privacy — Differential Privacy

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Dataset & Benchmark

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- Le Borgne, Y.-A., & Bontempi, G. (2022). Reproducible machine learning for credit card fraud detection — Practical handbook. Université Libre de Bruxelles. https://github.com/Fraud-Detection-Handbook

Literature Review Methodology

- Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, M. M., Li, T., Loder, E. W., Mayo-Wilson, E., McDonald, S., … Moher, D. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ*, 372, n71. https://doi.org/10.1136/bmj.n71

Evaluation Metrics

- Lebichot, B., Verhelst, T., Le Borgne, Y.-A., He-Guelton, L., Oblé, F., & Bontempi, G. (2022). Transfer learning strategies for credit card fraud detection. *IEEE Access*, 9, 146256–146268. https://doi.org/10.1109/ACCESS.2021.3122263

Non-IID Diagram (slide citation)

- Zhao, Y., Li, M., Lai, L., Suda, N., Civin, D., & Chandra, V. (2018). Federated learning with non-IID data. *arXiv preprint*. https://doi.org/10.48550/arXiv.1806.00582

Q&A